

SCOPING REVIEW PROTOCOL

Energy performance measurement models in continuous process industries under ISO 50001/50006

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Title	Energy performance measurement models in continuous process industries under ISO 50001/50006: a scoping review protocol
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1. Background and Rationale

Energy performance measurement is a central requirement of ISO 50001:2018, the international standard for energy management systems (EnMS). Operationalising this requirement demands the definition of Energy Performance Indicators (EnPIs) and Energy Baselines (EnBs) — the core instruments through which organisations quantify and track energy performance improvement over time, as detailed in ISO 50006:2023.

Despite the growing adoption of ISO 50001 across industrial sectors, the academic literature reveals a structural gap: less than 6% of the literature that employs EnPI/EnB terminology operates explicitly within the ISO 50001/50006 normative framework. The two domains — methodological EnPI/EnB research and ISO 50001 implementation studies — largely coexist independently. This suggests that existing EnPI/EnB models may not have been systematically developed or validated against the specific requirements of the normative framework.

A further concern is contextual. The ISO 50001/50006 framework was designed to be applicable to any type of organisation. However, continuous process industries — such as petroleum refining, petrochemicals, cement, and pulp and paper — exhibit operational characteristics that challenge standard EnB assumptions. Predictable, non-stationary process behaviours, such as heat exchanger fouling or catalyst deactivation cycles, cause systematic changes in the energy-production relationship that are not attributable to inefficiency but rather to inherent process dynamics. Standard baseline methodologies based on historical averages may be unable to distinguish these structural patterns from genuine performance changes.

A scoping review is the appropriate methodology to map this field, given the heterogeneity of approaches and the absence of a systematic synthesis addressing the specific adequacy of these models for continuous process industries. This protocol follows the framework of Arksey and O'Malley (2005), updated by Levac et al. (2010), and reports according to the PRISMA-ScR checklist (Tricco et al., 2018).

2. Objective and Research Questions

The overall objective of this scoping review is to **evaluate the adequacy of energy performance measurement models for continuous process industries**, mapping the state of the art of EnPI and EnB models within the ISO 50001/50006 normative framework, characterising their distribution by type of organisation and application context, and identifying methodological adaptations that may be required for their application in continuous process industries.

The primary research question is:

"To what extent are the energy performance measurement models proposed in the literature to fulfil the requirements of the ISO 50001/50006 normative framework adequate for continuous process industries, and what methodological adaptations may be necessary for their application?"

The following sub-questions guide the review:

- SQ1: What energy performance measurement models exist in the literature within the scope of the ISO 50001/50006 normative framework, for what contexts were they developed and validated, and how are they distributed across different types of organisation?
- SQ2: What operational variability mechanisms specific to continuous process industries may render these models inadequate for continuous monitoring aligned with the reference standards?
- SQ3: How may EnPI and EnB models be adapted to address the operational variability of continuous processes within the requirements of the reference standards?
- SQ4: To what extent are the methodological recommendations, if identified, applicable and operationalisable in a real petroleum refinery context?

3. Eligibility Criteria (PCC Framework)

3.1 Population

Organisations implementing or intending to implement energy management systems with quantitative performance measurement. The scope encompasses continuous process industries (petroleum refining, petrochemicals, cement, pulp and paper, steel) and discrete manufacturing (automotive, electronics, consumer goods). Services, residential buildings, and non-industrial public sector organisations are not excluded from the search strategy — their inclusion enables bibliometric distribution analysis by sector — but are treated as contextual contrast and recorded separately in the PRISMA flow diagram.

3.2 Concept

Models for energy performance measurement: EnPIs, EnBs, normalisation methodologies, measurement and verification (M&V) frameworks, and continuous monitoring systems. The central focus is the methodological adequacy of these models for different industrial contexts — in particular, whether they were developed and validated primarily for discrete manufacturing and present limitations in continuous process industries.

3.3 Context

The ISO 50001/50006 normative framework as the reference structure for energy performance measurement. Empirical validation context: Sines petroleum refinery (Galp Energia), specifically the atmospheric crude distillation unit, with planned extension to other process units.

Inclusion and exclusion criteria are summarised in the table below:

Criterion	Inclusion	Exclusion
Source types	Peer-reviewed journal articles; conference proceedings; doctoral dissertations; grey literature from internationally recognised reference organisations (ISO, IEA, DOE, LBNL, CONCAWE, NRCan, IOGP, IPIECA, Solomon Associates)	Master's dissertations; opinion articles without empirical or methodological basis; predatory publications
Language	English only	All other languages
Publication period	1996–present for systematic database search (justified by IPMVP 1996 as first systematic international M&V framework); no lower limit for seminal documents and standards identified through snowballing	Documents published before 1996 (database search only)
Minimum content	Document must present, propose, evaluate or critique an EnPI and/or EnB model with sufficient methodological detail for data extraction	Documents that mention EnPI/EnB only incidentally without methodological content
Context	Any organisational context at title and abstract screening levels; contextual exclusion criteria applied at full text screening level only, to enable bibliometric distribution analysis by sector	Exclusive focus on: residential buildings without industrial component; government energy policy without organisational dimension; purely economic energy management without performance measurement; ISO 14001/9001 without relation to ISO 50001 or energy management. Applied at full text screening level; exclusions recorded in PRISMA flow diagram.
Grey literature	Identified through backward snowballing of included papers, where such documents appear as cited references, and authored by reference organisations (ISO, IEA, DOE, LBNL, CONCAWE, NRCan, IOGP, IPIECA, Solomon Associates); reported separately in PRISMA flow diagram.	Grey literature from non-reference sources without clear institutional authorship

4. Search Strategy

4.1 Information Sources

Systematic search conducted in two bibliographic databases: **Scopus** (Elsevier) and **Web of Science Core Collection** (Clarivate). Access via b-on institutional consortium (Nova University Lisbon). Search executed in February 2026. Grey literature from reference organisations searched via institutional websites. Snowballing (backward and forward) conducted on the final included corpus.

4.2 Search Strings

Two complementary strings were developed following a calibration process across both databases. The calibration tested five string configurations, from a fully restrictive AND between both conceptual blocks to each block in isolation, documenting volume and noise at each step. The final two-string strategy was selected to maximise recall while maintaining a manageable screening volume.

String A (normative corpus) captures the full ISO 50001/50006 literature without forcing co-occurrence with EnPI/EnB terminology — enabling retrieval of implementation cases where measurement models are applied implicitly.

String B (methodological corpus) captures EnPI/EnB methodological literature anchored in industrial/management context, eliminating noise from unrelated domains where 'EnB' is a shared acronym.

String	Query (Scopus syntax)
String A Normative corpus	<pre>TITLE-ABS-KEY("ISO 50001" OR "ISO 50006") AND PUBYEAR > 1995 AND DOCTYPE(ar OR cp OR re)</pre> WoS equivalent: replace TITLE-ABS-KEY with TS; PUBYEAR > 1995 with PY=(1996-2026); DOCTYPE with DT=(Article OR Proceedings Paper OR Review)
String B Methodological corpus	<pre>TITLE-ABS-KEY(("energy performance indicator*" OR "EnPI" OR "energy baseline*" OR "EnB" OR "energy performance measurement" OR "energy performance monitoring") AND ("industrial" OR "industry" OR "manufacturing" OR "facility" OR "organization" OR "management system" OR "energy management" OR "ISO 50001" OR "continuous process" OR "refin*"))</pre> AND PUBYEAR > 1995 AND DOCTYPE(ar OR cp OR re) WoS equivalent: replace TITLE-ABS-KEY with TS; PUBYEAR > 1995 with PY=(1996-2026); DOCTYPE with DT=(Article OR Proceedings Paper OR Review)

Temporal limit: PUBYEAR > 1995 in both strings, justified by the publication of the IPMVP in 1996 as the first systematic international framework for energy measurement and verification — the foundational milestone for the Concept of this review. Literature predating this milestone enters through snowballing of seminal documents, not through systematic database search.

4.3 Search Volumes

String	Scopus	WoS	Notes
String A (normative corpus)	468	263	ISO 50001/50006, open
String B (methodological corpus)	486	261	EnPI/EnB + industrial context
Total (with overlap)	954	524	Before deduplication
Estimated unique records	~1,000–1,100	(combined)	After global deduplication in Rayyan

4.4 Sensitivity Check

Five anchor papers were verified against the final strings to validate recall adequacy. Four of five anchor papers were captured by at least one string. The fifth (Karcher, 2015) was not retrieved because its content concerns organisational success factors for ISO 50001 implementation without EnPI/EnB methodology — outside the Concept of the PCC. It enters formally through backward snowballing.

5. Selection of Sources of Evidence

Screening is conducted in three sequential levels using Rayyan systematic review management software:

- Level 1 — Title screening: all retrieved records reviewed by title only. Exclusion criterion: clear irrelevance to energy management in organisational contexts (e.g. medicine, pure physics, chemistry without energy management component).
- Level 2 — Abstract screening: records passing Level 1 reviewed by abstract. Inclusion/exclusion criteria applied as defined in Section 3. Records where decision cannot be made from abstract proceed to Level 3.
- Level 3 — Full text screening: full text obtained and reviewed for final eligibility determination. Contextual exclusion criteria applied at this level. All exclusions recorded with reason for PRISMA flow diagram.
- Snowballing: backward snowballing (reference lists of included papers) and forward snowballing (citing articles via Scopus 'Cited by' function) conducted on the final included corpus. New candidates pass through the same three-level screening process. Grey literature from reference organisations (ISO, IEA, DOE, LBNL, CONCAWE, NRCan, IOGP, IPIECA, Solomon Associates) will be captured through backward snowballing where such documents appear as cited references in included papers; grey literature identified through this process will be screened against the same inclusion/exclusion criteria and reported separately in the PRISMA flow diagram.

Screening conducted by a single reviewer (B. Ribeiro). Single-reviewer process is acknowledged as a methodological limitation and declared as such in the review report.

6. Data Charting

Data will be extracted from each included paper using a standardised charting form aligned with the four sub-questions. All categorical variables admit multiple selection. The charting form will be piloted on five papers before full extraction to verify consistency and completeness.

Category	Variable	Values / Notes
General (all papers)	Paper ID; authors; year; title; source; document type	Alphanumeric / categorical
	Country / region	Free text
	Sector of activity	Free text
	Type of organisation	Continuous process / discrete manufacturing / services / buildings / mixed / not specified
	Explicit normative context	ISO 50001 / ISO 50006 / both / none / other

SQ1 Model mapping	Type of EnPI/EnB model proposed	Linear regression / multiple regression / physical model / ML / composite index / other [multiple selection allowed]
	Normalisation variables used	Free text [multiple selection allowed]
	Baseline construction method	Free text
	Type of validation	Empirical / simulation / theoretical / none [multiple selection allowed]
	Scale of application	Equipment / unit / facility / organisation
SQ2 Limitations	Paper identifies limitations for continuous process	Yes / No
	Specific limitations identified	Non-stationarity / fouling / catalyst deactivation / feedstock variability / other [multiple selection allowed]
	Paper proposes treatment mechanism	Yes / No
SQ3 Adaptations	Paper proposes methodological adaptations for continuous process	Yes / No
	Type of adaptation proposed	Dynamic time window / segmented baseline / hybrid physical model / other [multiple selection allowed]
	Normative framing of adaptations	ISO 50001/50006 explicit / implicit / absent
SQ4 Refinery applicability	Empirical application in refinery or petrochemical	Yes / No
	Data type used	Continuous time series / audit data / simulation
	Performance metrics reported	Free text
	Implementation barriers identified	Free text [multiple selection allowed]

7. Presentation and Synthesis of Results

Results will be presented through:

- PRISMA-ScR flow diagram documenting records retrieved, deduplicated, screened at each level, and included/excluded with reasons.
- Descriptive bibliometric analysis: distribution of included papers by year, country, sector, organisation type, and normative context — addressing SQ1.
- Narrative synthesis organised thematically by sub-question: model typology (SQ1), identified limitations for continuous process (SQ2), proposed adaptations (SQ3), and refinery applicability (SQ4).
- Summary tables characterising the included corpus by key charting variables.

The review does not include meta-analysis or quantitative evidence synthesis.

8. Anticipated Limitations

- Single-reviewer screening and charting — potential for selection bias, mitigated by transparent documentation of decisions.
- English-language restriction — may exclude relevant literature published in other languages, particularly from non-Anglophone industrial regions.
- Grey literature coverage is partial — limited to documents from the reference organisations listed in Section 3.
- Temporal limit of 1996 for systematic search — pre-IPMVP literature on energy intensity indicators not captured through database search.

9. Methodological References

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